

CSE 5825: Probabilistic Graph Analysis

Homework 1

Problem 1 (10 points)

Prove

$$p(x, y \mid z) = p(x \mid z) p(y \mid x, z)$$

and also

$$p(x \mid y, z) = \frac{p(y \mid x, z) p(x \mid z)}{p(y \mid z)}.$$

First identity. Apply the ordinary product rule, $p(u, v) = p(u \mid v) p(v)$, inside the world where z is already given – i.e. treat $p(\cdot \mid z)$ as an ordinary distribution and apply the product rule to x and y within it, conditioning on x first:

$$p(x, y \mid z) = p(x \mid z) p(y \mid x, z).$$

Second identity. Apply the product rule again to the same quantity $p(x, y \mid z)$, but conditioning on y first instead:

$$p(x, y \mid z) = p(y \mid z) p(x \mid y, z).$$

The left-hand sides of the two expansions are identical, so the right-hand sides must be equal to each other:

$$p(x \mid z) p(y \mid x, z) = p(y \mid z) p(x \mid y, z).$$

Dividing both sides by $p(y \mid z)$ (assuming $p(y \mid z) \neq 0$) and solving for $p(x \mid y, z)$ gives

$$p(x \mid y, z) = \frac{p(y \mid x, z) p(x \mid z)}{p(y \mid z)},$$

which is exactly the second identity. ■

Problem 2 (15 points)

A college has three departments, A, B, and C, with female percentages 35%, 20%, and 55% respectively. A department is chosen at random, then a student is chosen at random from that department.

(a) $P(\text{female})$

(b) $P(\text{Dept} = B \mid \text{female})$

(c) $P(\text{Dept} = A \mid \text{male})$

Problem 3 (20 points)

Consider $P(A, B, C, D)$ where A, B are binary and C, D have 3 values each.

(a) Parameters for the full joint

(b) Parameters assuming $P(A, B, C, D) = P(A)P(B | A)P(C | B)P(D | B)$

(c) Network diagram and independence questions

Is $A \perp B$? Is $A \perp B | C$?

Problem 4 (30 points)

Chest Clinic belief network. Variables: x = positive X-ray, d = dyspnea, e = either tuberculosis or lung cancer, t = tuberculosis, l = lung cancer, b = bronchitis, a = visited Asia, s = smoker.

(a) Conditional independence statements (true/false)

1. tuberculosis $\perp$ smoking | shortness of breath
2. lung cancer $\perp$ bronchitis | smoking
3. visit to Asia $\perp$ smoking | lung cancer
4. visit to Asia $\perp$ smoking | lung cancer, shortness of breath

(b) $p(d)$, $p(d | s = \text{tr})$, $p(d | s = \text{fa})$

Computed via code (see `chest_clinic_skeleton.ipynb`, submitted alongside this file). Factorization used:

$$p(a, t, s, l, b, e, x, d) = p(a) p(t | a) p(s) p(l | s) p(b | s) p(e | t, l) p(x | e) p(d | e, b).$$

Skeleton of the computation (fill in CPT values and complete the marked steps in the notebook, then transcribe the final numbers here):

$$\begin{aligned} p(d) &= \\ p(d | s = \text{tr}) &= \\ p(d | s = \text{fa}) &= \end{aligned}$$

Problem 5 (15 points)

Consider $p(x, y, w, z) = p(z | w) p(w | x, y) p(x) p(y)$.

(a) Network diagram

(b) $p(x | z)$

(c) $p(y | z)$

(d) Condition for $x \perp y | z$

Problem 6 (15 points)

Graph: $A \rightarrow C$, $A \rightarrow E$, $B \rightarrow C$, $B \rightarrow D$, $C \rightarrow E$.

Algorithm 1 Marginal and conditional queries on the Chest Clinic network

```
1: procedure JOINTPROB( $a, t, s, l, b, e, x, d$ )
2:   return  $p(a) \cdot p(t \mid a) \cdot p(s) \cdot p(l \mid s) \cdot p(b \mid s) \cdot p(e \mid t, l) \cdot p(x \mid e) \cdot p(d \mid e, b)$ 
3: end procedure
4: procedure MARGINALD
5:    $p_d[\text{tr}] \leftarrow 0, \quad p_d[\text{fa}] \leftarrow 0$ 
6:   for each assignment of  $a, t, s, l, b, e, x, d$  do
7:      $p_d[d] \leftarrow p_d[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
8:   end for
9:   return  $p_d$ 
10: end procedure
11: procedure CONDITIONALDGIVENS( $s^*$ )
12:    $u[\text{tr}] \leftarrow 0, \quad u[\text{fa}] \leftarrow 0$ 
13:   for each assignment of  $a, t, s, l, b, e, x, d$  with  $s = s^*$  do
14:      $u[d] \leftarrow u[d] + \text{JOINTPROB}(a, t, s, l, b, e, x, d)$ 
15:   end for
16:   return  $u$  normalized so  $u[\text{tr}] + u[\text{fa}] = 1$ 
17: end procedure
```

- (a) Independent pairs, no observations
- (b) Independent pairs given E
- (c) Decomposition of $P(A, B, C, E)$
- (d) Parameters for CPD of E (A, B, D binary; C, E three values)